



EXAM PAPERS PRACTICE

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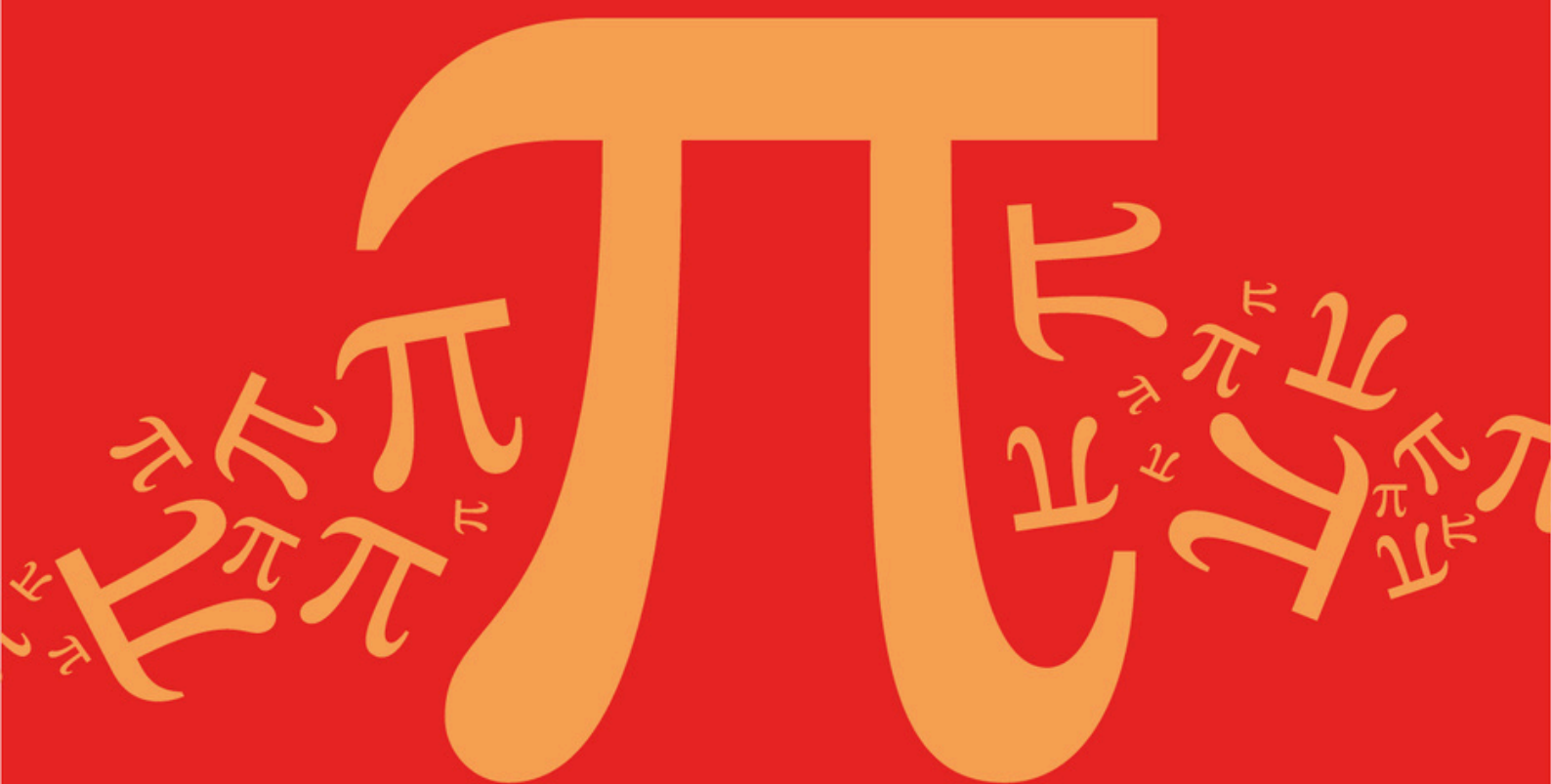
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AQA A Level Further Mathematics (7367)



Topic

B: Complex Numbers

Sub topic

B10: Nth roots

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic **B: Complex Numbers**
Sub topic **B10: Nth roots**

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

Solve the equation $z^3 = i$, giving your answers in the form $e^{i\theta}$, where $-\pi < \theta \leq \pi$

(Total 4 marks)

Q2.

(a) Express $-4 + 4\sqrt{3}i$ in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(3)

(b) (i) Solve the equation $z^3 = -4 + 4\sqrt{3}i$, giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(4)

(ii) The roots of the equation $z^3 = -4 + 4\sqrt{3}i$ are represented by the points P , Q and R on an Argand diagram.

Find the area of the triangle PQR , giving your answer in the form $k\sqrt{3}$, where k is an integer.

(3)

(c) By considering the roots of the equation $z^3 = -4 + 4\sqrt{3}i$, show that

$$\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9} = 0$$

(4)

(Total 14 marks)

Q3.

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(a) Write down the five roots of the equation $z^5 = 1$, giving your answers in the form $e^{i\theta}$, where $-\pi < \theta \leq \pi$.

(1)

(b) Hence find the four linear factors of

$$z^4 + z^3 + z^2 + z + 1$$

(3)

(c) Deduce that

$$z^2 + z + 1 + z^{-1} + z^{-2} = \left(z - 2\cos \frac{2\pi}{5} + z^{-1}\right) \left(z - 2\cos \frac{4\pi}{5} + z^{-1}\right)$$

(4)

(d) Use the substitution $z + z^{-1} = w$ to show that $\cos \frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$.

(6)

(Total 14 marks)

Q4.

- (a) Express in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$:

(i) $4(1 + i\sqrt{3})$;

(ii) $4(1 - i\sqrt{3})$.

(3)

- (b) The complex number z satisfies the equation

$$(z^3 - 4)^2 = -48$$

Show that $z^3 = 4 \pm 4\sqrt{3}i$.

(2)

- (c) (i) Solve the equation

$$(z^3 - 4)^2 = -48$$

giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(5)

- (ii) Illustrate the roots on an Argand diagram.

(3)

- (d) (i) Explain why the sum of the roots of the equation

$$(z^3 - 4)^2 = -48$$

is zero.

(1)

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- (ii) Deduce that $\cos \frac{\pi}{9} + \cos \frac{3\pi}{9} + \cos \frac{5\pi}{9} + \cos \frac{7\pi}{9} = \frac{1}{2}$.

(3)

(Total 17 marks)

Q5.

- (a) (i) Show that $\omega = e^{\frac{2\pi i}{7}}$ is a root of the equation $z^7 = 1$.

(1)

- (ii) Write down the five other non-real roots in terms of ω .

(2)

- (b) Show that

$$1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 + \omega^6 = 0$$

(2)

- (c) Show that:

$$(i) \quad \omega^2 + \omega^5 = 2 \cos \frac{4\pi}{7}; \quad (3)$$

$$(ii) \quad \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}. \quad (4)$$

(Total 12 marks)

Q6.

(a) Show that

$$(z^4 - e^{i\theta})(z^4 - e^{-i\theta}) = z^8 - 2z^4 \cos \theta + 1 \quad (2)$$

(b) Hence solve the equation

$$z^8 - z^4 + 1 = 0$$

giving your answers in the form $e^{i\phi}$, where $-\pi < \phi \leq \pi$. (6)

(c) Indicate the roots on an Argand diagram.

(3)
(Total 11 marks)

Q7.

Given that $z = 2e^{i\frac{\pi}{12}}$ satisfies the equation

$$z^4 = a(1 + \sqrt{3}i)$$

where a is real:

(a) find the value of a ;

(3)

(b) find the other three roots of this equation, giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(5)
(Total 8 marks)

Q8.

(a) Express $4 + 4i$ in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(3)

(b) Solve the equation

$$z^5 = 4 + 4i$$

giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(5)
(Total 8 marks)

Q9.

- (a) Find the three roots of $z^3 = 1$, giving the non-real roots in the form $e^{i\theta}$, where $-\pi < \theta \leq \pi$.

(2)

- (b) Given that ω is one of the non-real roots of $z^3 = 1$, show that

$$1 + \omega + \omega^2 = 0$$

(2)

- (c) By using the result in part (b), or otherwise, show that:

(i) $\frac{\omega}{\omega + 1} = -\frac{1}{\omega};$

(2)

(ii) $\frac{\omega^2}{\omega^2 + 1} = -\omega;$

(1)

(iii) $\left(\frac{\omega}{\omega + 1}\right)^k + \left(\frac{\omega^2}{\omega^2 + 1}\right)^k = (-1)^k 2\cos\frac{2}{3}k\pi,$ where k is an integer.

(5)

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Q10.

- (a) (i) Given that $z^6 - 4z^3 + 8 = 0$, show that $z^3 = 2 \pm 2i$.

(2)

- (ii) Hence solve the equation

$$z^6 - 4z^3 + 8 = 0$$

giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(6)

- (b) Show that, for any real values of k and θ ,

$$(z - ke^{i\theta})(z - ke^{-i\theta}) = z^2 - 2kz\cos\theta + k^2$$

(2)

- (c) Express $z^6 - 4z^3 + 8$ as the product of three quadratic factors with real coefficients.

(3)

(Total 13 marks)

Q11.

- (a) Find the six roots of the equation $z^6 = 1$, giving your answers in the form $e^{i\phi}$, where $-\pi < \phi \leq \pi$.

(3)

- (b) It is given that $w = e^{i\theta}$, where $\theta \neq n\pi$.

(i) Show that $\frac{w^2 - 1}{w} = 2i \sin \theta$.

(2)

(ii) Show that $\frac{w}{w^2 - 1} = -\frac{i}{2 \sin \theta}$.

(2)

(iii) Show that $\frac{2i}{w^2 - 1} = \cot \theta - i$.

(3)

(iv) Given that $z = \cot \theta - i$, show that $z + 2i = zw^2$.

(2)

- (c) (i) Explain why the equation

$$(z + 2i)^6 = z^6$$

has five roots.

(1)

- (ii) Find the five roots of the equation

$$(z + 2i)^6 = z^6$$

giving your answers in the form $a + ib$.

(4)

(Total 17 marks)